Overview

This assessment is designed to synthesise the knowledge and skills acquired in the operationalisation of Generative AI technologies within a business context. You will step into the role of an AI consultant for **Modern Furniture Co.**, a business facing a variety of operational challenges.

Your task is to design, propose, and create a proof-of-concept for a generative AI-powered **Analytics Agent**. This agent will enable the management of Modern Furniture Co. to interact with their business data through natural language, ask critical questions, and receive data-driven insights to solve key operational problems.

Students are required to analyse the provided **Case Study** and are encouraged to examine the **Advanced Furniture Manufacturing Simulator** to gain a deeper understanding of the business dynamics. The goal is to reimagine how business analytics is done in the age of AI and to develop an agent that helps Modern Furniture Co. address its specific challenges.

Purpose

The purpose of this assessment is twofold:

1. To cultivate your skills in identifying and analysing a business's operational challenges and data-driven needs from a case study.
2. To provide you with hands-on experience in designing and building a functional AI agent that connects to a dataset to solve real-world problems.

This exercise is designed to deepen your understanding of how AI can be effectively integrated into business operations to drive efficiency and strategic decision-making.

Learning Outcomes

The targeted Course Learning Outcomes (CLOs) for this assessment are:

- **CLO1**: Evaluate and discuss new and emerging AI and analytics applications.
- **CLO2**: Critically review AI and analytics technologies, and how they can be used to achieve operational efficiencies in business.
- **CLO4**: Evaluate the risks and ethical concerns associated with AI and Analytics products.
- **CLO5**: Recommend design patterns for common AI and analytics solutions for business and analyse the typical life-cycle of AI and analytics projects.

Marking criteria

You will be marked on your:

- Research and analysis (10%)
- Business requirements (20%)
- Conceptual design (20%)
- Data integration plan (10%)
- Proof of concept (20%)
- Conclusions (10%)
- Format and professionalism (10%)

A more detailed rubric can be found below.

Scenario: AI-Powered Analytics at Modern Furniture Co.

Project Title: The Modern Furniture Co. AI Analytics Agent

Introduction:

Modern Furniture Co. is a custom furniture manufacturer looking to leverage its operational data to become more profitable and efficient. The company uses a sophisticated business simulator that generates rich data across all facets of its operations, from sales and finance to production and human resources. However, the management team lacks the tools to easily query this data to get timely, actionable insights.

Your task is to propose and build a proof-of-concept for an AI Analytics Agent. This agent will act as an intelligent assistant, allowing a manager to ask questions in plain English like:

- "What was our biggest production bottleneck last quarter?"
- "Show me the impact of employee resignations on our on-time delivery rate."
- "Which product type is the most profitable?"

The agent should process these queries, analyse the relevant simulator data, and provide answers that may include text summaries, key metrics, and visualisations.

Objectives:

- To provide an interactive and intuitive way for non-technical business users to explore operational data.
- To deliver timely and relevant insights to address the core challenges identified in the case study (e.g., demand forecasting, production efficiency, cost control, reputation management).
- To leverage generative AI technology to understand user intent and generate insightful responses based on the provided datasets.
- To assist management in making strategic, data-driven decisions to improve the overall performance of the business.

Data Source:

You will used this dataset

Download dataset

generated by the Advanced Furniture Manufacturing Simulator. This dataset includes multiple CSV files such as a Financial Ledger, All Orders, HR Events Log, Inventory Log, etc. Your AI agent must be designed to integrate with and analyse this data.

Instructions

You will develop a slide deck and a short demonstration video for this assessment.

Slide Deck Format (10 slides maximum):

1. **Project Overview:** Provide a brief overview of your proposed AI Analytics Agent for Modern Furniture Co., explaining its purpose and value proposition.
2. **Business Problem Analysis:** Based on the case study, identify and analyse the key operational challenges and business questions facing Modern Furniture Co. Prioritise which problems your AI agent will focus on solving.
3. **Business Requirements:** Define the specific functional requirements for your agent. What specific questions will it be able to answer? What data will it need? What will the outputs be?
4. **Conceptual Design:** Create a conceptual design for your agent. Illustrate the user interaction flow (e.g., how a user asks a question and receives a response). You can use sketches, wireframes, or digital tools.
5. **Data Integration Plan:** Detail your plan for integrating and using the data from the simulator. Specify which CSV files are needed to answer your identified business questions and how the data will be processed.
6. **Proof of Concept (POC) Demonstration:** Embed or link to your screen-recording video demonstrating your POC. This video should show your agent in action, answering at least 2-3 distinct business questions using the provided data. Explain the technology you used (e.g., Google AI Studio, Microsoft Copilot Studio, or another platform).
7. **Testing and Feedback:** Briefly describe how you tested your agent and what the results were. If you gathered feedback from peers, summarise it here.
8. **Conclusions & Recommendations:** Conclude with a summary of the project, its limitations, and recommendations for future development to make the agent even more powerful for Modern Furniture Co.

Submission:

- Submit your slide deck as a single file.
- Submit your screen-recording video of your proof-of-concept via Canvas Studio or a linked video in your slides.

Resources

Examples and supporting resources will be provided as you work through weekly challenges.

Assessment 2 rubric

Criteria	Ratings	Pts
This criterion is linked to a learning outcome C1 (10%) Business Problem Analysis	**10 to >7.99 PtsHigh Distinction: 100-80% - Highly developed / Excellent**Comprehensive and insightful analysis of the case study. Identifies and clearly articulates critical business challenges and opportunities for AI intervention. **7.99 to >6.99 PtsDistinction: 79%-70% - Developed / Very good**Good analysis of the case study, identifying key business challenges with relevant connections to AI solutions. **6.99 to >5.99 PtsCredit: 69%-60%: Emerging / Competent**Adequate analysis of the case study with a basic understanding of the main business challenges. **5.99 to >4.99 PtsPass / Satisfactory: 59%-50%**Limited analysis of the case study, with some challenges identified but lacking depth or clarity. **4.99 to >0.0 PtsNot satisfactory: Less than 50% Fails to meet standards**Fails to meaningfully analyse the case study or identify relevant business problems. **0 PtsDid not submit**Did not submit.	10 pts

This criterion is linked to a learning outcome C2 (20%) Business Requirements	**20 to >15.98 PtsHD**Clear, well-defined, and innovative business requirements that directly address the core problems of Modern Furniture Co. Functionalities are specific, measurable, and highly relevant. **15.98 to >13.98 PtsDI**Strong and clear business requirements that address the key needs of the business, with minor areas for improvement in specificity or innovation. **13.98 to >11.98 PtsCR**Adequate business requirements that align with the case study, but may lack detail, uniqueness, or depth. **11.98 to >9.98 PtsPA**Limited or vague business requirements with unclear alignment to the identified business problems. **9.98 to >0.0 PtsNN**No clear or relevant business requirements identified. **0 PtsDNS**Did not submit.	20 pts

This criterion is linked to a learning outcome #AoL5 Expert C3 (20%) Conceptual Design	**20 to >15.96 PtsHD**Exceptional and innovative design with a clear, intuitive, and highly effective user interaction flow. The design is well-reasoned and user-centric. **15.96 to >13.96 PtsDI**Good design with a clear interaction flow and useful features. The design is usable and engaging. **13.96 to >11.96 PtsCR**Satisfactory design with a generally clear interaction flow, but usability and effectiveness could be improved. **11.96 to >9.96 PtsPA**Basic design with an unclear or ineffective interaction flow. Usability is limited. **9.96 to >0.0 PtsNN**No clear or usable conceptual design presented. **0 PtsDNS**Did not submit.	20 pts
This criterion is linked to a learning outcome C4 (10%) Data Integration Plan	**10 to >7.99 PtsHD**Comprehensive and well-defined data integration plan. Clearly identifies all necessary data sources from the simulator dataset and details robust methods for processing and analysis. **7.99 to >6.99 PtsDI**Good integration plan with appropriate data sources and processing methods, though some areas could be more detailed. **6.99 to >5.99 PtsCR**Adequate data integration plan, but may be lacking in detail or clarity on data processing methods. **5.99 to >4.99 PtsPA**Limited data integration plan with unclear or missing sources and processing methods. **4.99 to >0.0 PtsNN**No viable data integration plan provided. **0 PtsDNS**Did not submit.	10 pts

This criterion is linked to a learning outcome #AoL4 Digitally Adept C5 (20%) Proof of Concept	**20 to >16.0 PtsHD**High-quality, functional prototype that effectively implements all key functionalities. The agent handles interactions smoothly, provides accurate, data-driven answers, and is engaging to use. **16 to >14.0 PtsDI**Functional prototype with most key functionalities implemented well, though some features may need refinement. **14 to >12.0 PtsCR**A working prototype that implements some core functionalities but may be limited in its interaction or the depth of its analytical capabilities. **12 to >10.0 PtsPA**Prototype with minimal functionality that struggles to be effective or interactive. **10 to >0.0 PtsNN**No functional prototype developed. **0 PtsDNS**No functional prototype developed.	20 pts

This criterion is linked to a learning outcome C6 (10%) Conclusions	**10 to >7.99 PtsHigh Distinction: 100-80% - Highly developed / Excellent**Comprehensive conclusions that critically reflect on the project's success and limitations, with clear, insightful, and actionable recommendations for future improvements. **7.99 to >6.99 PtsDistinction: 79%-70% - Developed / Very good**Clear conclusions with good recommendations for improvements and future applications. **6.99 to >5.99 PtsCredit: 69%-60%: Emerging / Competent**Adequate conclusions with some recommendations for improvement, but limited critical analysis. **5.99 to >4.99 PtsPass / Satisfactory: 59%-50%**Basic conclusions with vague or minimal recommendations. **4.99 to >0.0 PtsNot satisfactory: Less than 50% Fails to meet standards**No meaningful conclusions or recommendations provided. **0 PtsDid not submit**No meaningful conclusions or recommendations provided.	10 pts

This criterion is linked to a learning outcome C7 (10%) Format and Professionalism	**10 to >7.99 PtsHigh Distinction: 100-80% - Highly developed / Excellent**Impeccable presentation in terms of structure, clarity, and professional appearance. All slides and the video are clear, concise, and well-designed. **7.99 to >6.99 PtsDistinction: 79%-70% - Developed / Very good**Well-structured and clear presentation with minor issues in clarity or design. **6.99 to >5.99 PtsCredit: 69%-60%: Emerging / Competent**Adequate presentation, though some slides or parts of the video may lack clarity or consistency in design. **5.99 to >4.99 PtsPass / Satisfactory: 59%-50%**Presentation lacks clarity, with multiple issues in structure, formatting, or professionalism. **4.99 to >0.0 PtsNot satisfactory: Less than 50% Fails to meet standards**Presentation is unprofessional, unclear, and poorly structured. **0 PtsDid not submit**Did not submit.	10 pts

Total points: 100